2. Used in Lamp Dimmer Circuit.
3. Used in Heater Control Circuit.
4. Used in Motor Speed Control.

6.13. TRIAC

TRIAC is a three terminal semiconductor switching device. They are MT_1, MT_2 and gate. Here, the gate terminal is used to control the *AC* in a load. TRIAC is a short version of TRIODE AC switch. The flow of current in TRIAC is bi-directional that means current can flow in both directions.

Construction: The structure and symbol of TRIAC is shown in the Figure 6.21. It comprises of two SCRs connected in the anti-parallel direction. It acts as a switch for both the directions. From the diagram we can understand that the MT_1 and gate terminals are close to each other. The gate provides control over conduction in either direction.

Working Principle: Figure 6.22 shows the TRIAC circuit; here the A.C supply is connected across the main terminals (MT_1&MT_2) of TRIAC through a load resistor R_L. The gate circuit consists of battery, a variable resistor and switch. The TRIAC operation depends upon the polarity of voltage across its main terminals and gate terminal.

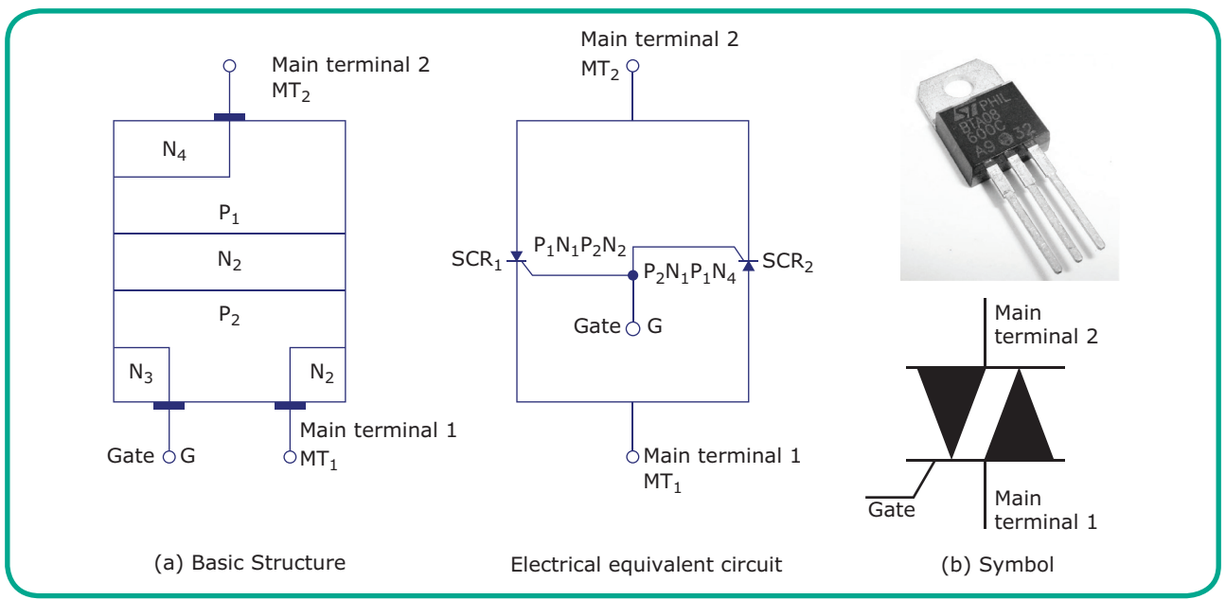


Figure 6.21 TRIAC

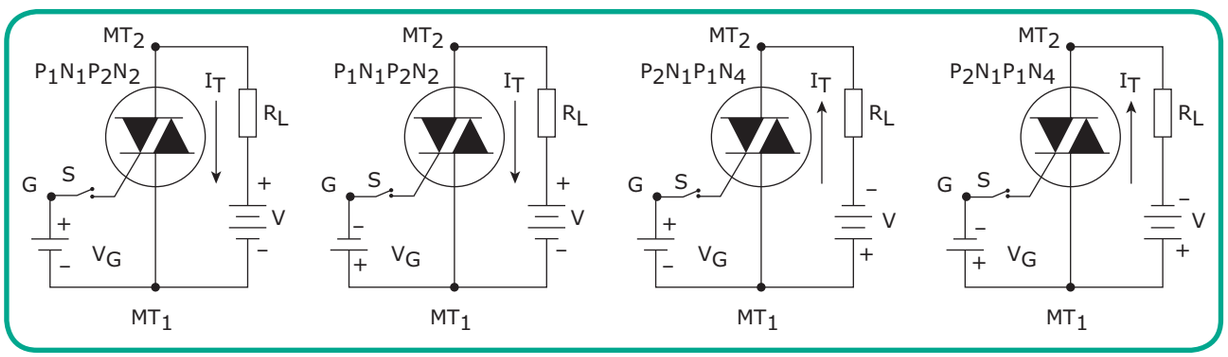


Figure 6.22 TRIAC Biasing Circuit

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